

Context-Gated Multi-Channel Hebbian Plasticity Cannot Bootstrap Channel Separation

A negative-result technical report • Revision 2

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Abstract

We investigate an architectural idea: augment the biologically-inspired *Dragon Hatchling* (BDH) language-model architecture so that each synaptic connection carries *multiple* weight channels, with a learned, context-dependent gate selecting or blending channels per token. The motivation is that one network might then express several distinct “modes” of working memory—analogous to a brain synapse using multiple neurotransmitters—without separate mixture-of-experts modules. We derive the mechanism, implement it exactly (verifying our recurrent implementation against the reference architecture to machine precision), and then test whether the channels do anything useful. They do not. Across five experiments of escalating directness—a cross-context recall task, an interference task where channel separation is provably beneficial, an auxiliary-loss sweep explicitly rewarding separation, a hard discrete-routing variant with a stability probe, and an asymmetric variant that decouples the write gate from the read gate—the gate consistently collapses to permissive, near-uniform routing and never separates the channels, capturing none of a 50-accuracy-point margin that a hand-set gate demonstrably claims. The final experiment yields a *structural* explanation that corrects the mechanism proposed in Revision 1 of this report. Because the write gate is a normalized distribution over channels, the channel-summed state is invariant to write routing; consequently, under a uniform read gate the write routing cancels *exactly*, and the gradient of the loss with respect to the write-gate parameters is identically zero. We verify this cancellation numerically to 9.4×10^{-16} . Training therefore begins in a deadlock: the write gate receives no signal about how to route because the read gate cannot see routing, and the read gate has no separated memories worth attending to because the write gate has not routed. Neither side can move first. This refutes the constructive fix proposed in Revision 1—decoupling the write from the routing—which we implemented and measured at zero lift. Any viable intervention must act on the read gate, not the write gate alone.

1 Background

1.1 The Dragon Hatchling (BDH) architecture

BDH [1] is a recently proposed language-model architecture designed as a bridge between the Transformer and biological models of the brain. Its distinguishing feature is that the model’s *working memory during inference* is stored in synaptic state that is updated by a Hebbian learning rule—“neurons that fire together wire together”—rather than in a fixed-length activation vector as in a recurrent network, or a growing key–value cache as in a Transformer.

The paper defines two related forms. The full **BDH** stores state as an $n \times n$ synaptic matrix σ , where n is a (large) number of neurons. The GPU-friendly **BDH-GPU**, which is the version implemented in the public code release, stores a mathematically equivalent but compressed state $\rho \in \mathbb{R}^{n \times d}$ with $d \ll n$, related to the full synaptic state by $\rho = E\sigma$ for an encoder matrix E . The

core recurrence, following equation (8) of the BDH paper [1], is

$$\rho_{t,l} = (\rho_{t-1,l} + \text{LN}(E y_{t,l-1}) x_{t,l}^\top) U, \quad (1)$$

$$x_{t,l} = x_{t-1,l} + (D_x \text{LN}(E y_{t,l-1}))^+, \quad (2)$$

$$y_{t,l} = (D_y \text{LN}(\rho_{t-1,l} x_{t,l}))^+ \odot x_{t,l}, \quad (3)$$

where $x, y \in \mathbb{R}^n$ are sparse, positive neuron activations, $v^* \in \mathbb{R}^d$ is a compressed value vector, $(\cdot)^+$ is a ReLU, $\odot$ is elementwise product, LN is a parameter-free LayerNorm, and U is a diagonal or block-diagonal matrix implementing positional decay/rotation (ALiBi or RoPE). The trainable parameters are the three matrices E, D_x, D_y .

The essential fact for this report is that the linear-attention operation in the public code is *identical* to a token-by-token Hebbian accumulation of the state matrix. The state at time t can be written explicitly (equation (16) of [1]) as a sum of outer products over the past:

$$\rho_{t-1,l} = \sum_{\tau < t} v_{\tau,l-1}^* x_{\tau,l}^\top U^{t-\tau}, \quad a_{t,l}^* = \rho_{t-1,l} x_{t,l}. \quad (4)$$

Reading memory is $x_t^\top$ applied to the accumulated state; writing memory is adding the outer product $v_t^* \otimes x_t$ at each step. This outer-product write *is* the Hebbian rule.

1.2 The proposed extension: multiple channels per synapse

The idea studied here is to give the synaptic state k independent channels. Concretely we maintain a state tensor $S \in \mathbb{R}^{k \times n \times d}$, a learned gate projection $W_g \in \mathbb{R}^{d \times k}$, and for each token a gate distribution

$$g_t = \text{softmax}(v_t^* W_g) \in \Delta^{k-1}, \quad (5)$$

a probability vector over the k channels. The Hebbian write deposits each token’s outer product into the channels in proportion to the gate,

$$S^{(c)} += g_t[c] (x_t \otimes v_t^*), \quad c = 1, \dots, k, \quad (6)$$

and the read blends the per-channel outputs by the same gate,

$$a_t^* = \sum_c g_t[c] (x_t^\top S^{(c)}). \quad (7)$$

Working through the algebra, the effective attention score between a query token t and a past key token τ becomes

$$\underbrace{(x_t \cdot x_\tau)}_{\text{content match}} \times \underbrace{(g_t \cdot g_\tau)}_{\text{context-gate match}}. \quad (8)$$

The content term is ordinary BDH attention; the new gate term modulates it. The design intent is that tokens sharing a context (similar g) can read each other’s memory, while tokens in different contexts (near-orthogonal g) are isolated into separate channels. When $k = 1$ the gate is trivially the scalar 1 and the mechanism reduces exactly to single-channel BDH.

1.3 Relationship to prior work

The blending mechanism itself is not novel. Computing a weighted combination of a bank of weight matrices with input-dependent coefficients is the core of CondConv [2], and closely related ideas appear as parameter superposition [3] and context-dependent gating [4]. All of these fall under the umbrella of hypernetworks / conditional weight generation. What is specific to the present work—and what makes the negative result below non-obvious—is the combination of context-dependent routing with *inference-time Hebbian plasticity*: unlike CondConv, whose weight banks are frozen, here each channel carries a state that *evolves during the forward pass*. It is precisely this interaction that produces the failure we document. Finally, the collapse we observe is an instance of the well-known *router collapse* phenomenon in mixture-of-experts models [7]; our contribution is to identify a novel and specific driver of it.

2 Implementation and verification

Before testing whether the mechanism helps, we verified that our implementation computes what the equations specify. This matters because the recurrent (token-by-token) form used for the experiments must match the parallel (token-parallel) form that the reference code and the paper’s Appendix E define.

We implemented both forms and checked their outputs agree. All checks use double precision; the reference threshold for “machine precision” is $\sim 10^{-15}$.

Verification check	Max abs. difference
Single-channel recurrent vs. parallel reference (Appendix E)	1.78×10^{-15}
Multi-channel ($k=3$) parallel vs. recurrent	8.88×10^{-16}
$k=1$ reduction to single-channel (parallel path)	0 (bitwise identical)
$k=1$ reduction to single-channel (recurrent path)	1.78×10^{-15}

Table 1: The recurrent multi-channel implementation matches the reference architecture to machine precision, and reduces exactly to single-channel BDH at $k = 1$. The bitwise-identical parallel reduction follows because at $k = 1$ the softmax returns exactly 1.0 and the gate outer product is an all-ones matrix, leaving the computation path algebraically unchanged.

Two limitations of the verification should be stated plainly. First, the equivalence was checked with $U = I$ (no positional decay or rotation); this is a valid instance of the paper’s equations and isolates the Hebbian identity cleanly, but the folding of RoPE/ALiBi into the recurrence is structurally standard rather than bit-verified here. Second, the full $n \times n$ synaptic matrix σ is only materialized at small n for analysis; at scale one trains on the compressed form ρ and recovers slices of σ via equation (4) only for interpretability. Neither limitation affects the conclusions below, which concern the compressed, trainable form.

Experiment 5 (Section 3.5) uses hard write routing, which admits no unconditional parallel form; the verification strategy for that variant is given separately in Section 5.

3 Experiments

All experiments use a small model ($d = 32$, $n = 64$, byte-level vocabulary) so that each run completes in minutes on CPU, and every result is reported as mean $\pm$ standard deviation over 3 random seeds. Each experiment isolates the loss to the specific token positions where the effect under test would appear (the query positions), rather than averaging over whole sequences.

3.1 Experiment 1: cross-context recall

We first asked whether the channels separate when nothing rewards separation. The task presents key-value associations under one context, then a query under a *different* context that can only be answered by recalling the earlier association. If the channels isolated the two contexts, the query could not read the earlier memory—so a working model here *must* keep memory shared.

Result. All conditions—single-channel, multi-channel with a learned gate, and multi-channel with a forced-uniform gate—achieved essentially perfect query accuracy (loss ≈ 0.0002 , accuracy = 1.000), indistinguishable at every checkpoint. Inspecting the learned gate showed it did *not* route the two contexts to different channels: the gate collapsed to near-uniform. The apparent conclusion that “isolation is free” is misleading, because the isolation mechanism never engaged. The correct reading is that end-to-end training drives the gate toward permissiveness, since separation could only hurt on this task. This motivated a task where separation *should* help.

3.2 Experiment 2: interference (separation is provably beneficial)

We constructed a task where a single shared memory is provably insufficient. Two interleaved streams use the *same* key tokens but assign them *conflicting* values (stream 1: $X \rightarrow 5$; stream 2: $X \rightarrow 9$, for the same key X). A per-stream context signal drives the gate. At query time, “ $X=?$ ” must return 5 under the stream-1 context and 9 under the stream-2 context. With a single Hebbian memory, both associations are written to the same synapses and average out—the model cannot exceed a hard ceiling. With separated channels, each stream’s association lives in its own state and both are recoverable. A sanity check confirmed that the analytical single-channel maximum is 50% accuracy.

Condition	Stream-1 acc	Stream-2 acc	Lift fraction
Single-channel baseline	0.508 ± 0.032	0.493 ± 0.022	— (floor)
MC $k=2$, learned gate	0.502 ± 0.027	0.490 ± 0.026	−0.01
MC $k=2$, <i>perfect</i> gate (hand-set)	1.000 ± 0.000	1.000 ± 0.000	1.00 (ceiling)
MC $k=2$, uniform gate	0.508 ± 0.032	0.493 ± 0.022	0.00 (floor)

Table 2: Interference task. The “lift fraction” is $(\text{acc} - \text{floor})/(\text{ceiling} - \text{floor})$: 1.0 means the condition matches perfect separation, 0.0 means it matches no separation. A hand-set gate solves the task perfectly (ceiling = 100%), but the learned gate captures −1% of that benefit and sits exactly on the floor.

Result. The hand-set “perfect gate” condition confirms that the architecture is fully *capable* of the task: with the right routing, accuracy is 100% where a single channel caps at 50%. Yet the *learned* gate captured none of this. The cosine similarity between the two streams’ gate vectors was 0.9993—the gate did not separate the streams at all, even though separation was the only path above 50%. The failure is not in the model’s representational capacity; it is in the learning dynamics.

3.3 Experiment 3: explicit auxiliary separation loss

We next added a loss term that directly rewards the gate for routing different-context tokens to different channels, scaled by how different their contexts are:

$$\mathcal{L}_{\text{aux}} = -\lambda \cdot \text{mean}_{i,j} \left[\|c_i - c_j\| (1 - \cos(g_i, g_j)) \right], \quad (9)$$

where c is the per-token context signal. We swept $\lambda \in \{0, 0.01, 0.1, 1.0\}$ —a hundredfold range.

Result. No value of λ produced separation or lift. The gate cosine stayed pinned near 1.0 across the entire sweep (ending at 0.9999 at $\lambda = 1.0$, marginally *more* collapsed than at $\lambda = 0$), and the lift fraction never left the floor (0.00 at best). Notably, the task loss did *not* degrade as λ increased—there was no over-forcing, meaning the auxiliary loss and the task loss were *not in tension*. The gate simply ignored a direct, strong incentive to separate. This indicates the collapse is not a shallow optimization artifact one hyperparameter away from a fix.

3.4 Experiment 4: hard discrete routing and a stability probe

Finally we replaced the soft gate blend with hard, discrete routing via a straight-through estimator: the forward pass commits to a single channel per token (a one-hot choice), while gradients flow through the underlying soft gate. Both the read and the Hebbian write use the hard choice. We ran two conditions: (1) standard random initialization, testing whether hard routing spontaneously separates; and (2) a gate *initialized* to the perfect separated solution, testing whether that solution is *stable* under training.

Condition	Stream-1 routing	Stream-2 routing	Lift
Discrete gate, standard init	ch0 0.41, ch1 0.59	ch0 0.52, ch1 0.48	−0.00
Discrete gate, <i>separated</i> init	ch0 0.64, ch1 0.36	ch0 0.30, ch1 0.70	−0.01

Table 3: Discrete-routing conditions. “Routing” is the fraction of each stream’s tokens sent to each channel; clean separation would read approximately (1.0, 0.0) for one stream and (0.0, 1.0) for the other. Standard init shows no stream/channel correspondence. Separated init—handed the perfect solution—drifts back toward mixing, and its lift returns to the floor.

Result. From standard initialization, discrete routing showed no reliable stream/channel correspondence and no lift—the same collapse as the soft case, reached by a different route. The decisive finding is condition 2: initialized to the perfect separated solution (the configuration that the hand-set gate of Experiment 2 proved achieves 100%), training *eroded* it, drifting from a clean split back toward mixing (0.64/0.36 and 0.30/0.70) with lift returning to −0.01. The separated solution is therefore not merely hard to *find*: it is not a stable fixed point of the training objective.

3.5 Experiment 5: asymmetric routing (write/read decoupling)

Revision 1 of this report concluded with a constructive proposal: since the router appeared to be penalized for withholding writes, one should *decouple the write from the routing* so that unrouted channels are not penalized for staying empty. Experiment 5 implements exactly that proposal and measures it.

We split the single gate into two. The Hebbian *write* commits to a single channel per token via a Gumbel-Softmax (Concrete) relaxation [5, 6] with an annealed temperature, rather than the straight-through estimator of Experiment 4, whose gradient estimation is known to be poor. The *read* continues to blend all channels with a soft gate. This is “hard write, soft read.” We tested both a shared gate projection W_g (write and read derived from the same logits) and fully separate projections. The implementation and the measured results are on branch `claude/bdh-growth-hebbian-inference-w90069` (commits 590f8e3, 054a910).

Result: the intervention engaged and still failed. This is not a case of a mechanism silently failing to activate. Mean write commitment reached 0.870 when sampled as during training (0.5 is uniform, 1.0 is one-hot), with a mean inter-channel logit gap of 0.80: the writes genuinely committed to single channels. What the committed write never did was *separate the streams*. At the end

Condition	Stream-1 loss	Stream-2 loss	Lift
1. Single-channel (floor)	0.6972 ± 0.0042	0.7004 ± 0.0029	—
2. Symmetric soft gate	0.6973 ± 0.0047	0.7005 ± 0.0023	−0.01
3. Asymmetric, shared W_g	0.6973 ± 0.0045	0.7003 ± 0.0027	0.00
3b. Asymmetric, separate W_g	0.6976 ± 0.0049	0.7013 ± 0.0026	−0.01
4. Perfect gate (ceiling)	0.0001	0.0001	1.00

Table 4: Asymmetric routing on the interference task. Both asymmetric variants are indistinguishable from the floor and from the symmetric gate that had already collapsed. A full 50 accuracy points were available (1.000 vs. 0.500) and none were claimed.

of training, stream-1 tokens wrote $[0.335, 0.665]$ and stream-2 tokens wrote $[0.448, 0.552]$ —both streams preferring the same channel. The separate- W_g variant shows the same pattern mirrored ($[0.522, 0.478]$ and $[0.639, 0.361]$, both favoring channel 0). There is no dead-channel collapse in either variant; the failure is *diffuse non-separation*, not degeneracy. The training trace shows separation never emerges at any checkpoint—routing is as unseparated at step 1200 as at step 0, and the read-gate cosine sits at 0.96–0.99 throughout, matching the symmetric condition’s 0.9993 almost exactly.

We checked the obvious objection that a deterministic evaluation might hide a working stochastic mechanism: sampled evaluation gives 0.481 across five independent draws, identical to the deterministic 0.481. This is not an evaluation artifact.

4 Why the mechanism fails: an exact cancellation

Pursuing the identical-across-draws result led to a structural explanation that is worth more than the negative result itself, and which *corrects* the mechanism proposed in Revision 1 of this report.

4.1 The channel sum is invariant to write routing

Write the asymmetric update with a normalized write gate $g_t^w \in \Delta^{k-1}$:

$$S_T^{(c)} = \sum_{t < T} g_t^w[c] (x_t \otimes v_t^*). \quad (10)$$

Summing over channels and exchanging the order of summation,

$$\sum_{c=1}^k S_T^{(c)} = \sum_{t < T} \left(\underbrace{\sum_{c=1}^k g_t^w[c]}_{=1} \right) (x_t \otimes v_t^*) = \sum_{t < T} (x_t \otimes v_t^*). \quad (11)$$

The channel-summed state is therefore *completely independent of the write routing*. This follows from normalization alone: routing does not change how much is written, only where, and the total is conserved.

4.2 Under a uniform read, the write routing cancels exactly

Now read with a uniform gate $g^r = (1/k)\mathbf{1}$:

$$a_t^* = \sum_{c=1}^k \frac{1}{k} (x_t^\top S^{(c)}) = \frac{1}{k} x_t^\top \sum_{c=1}^k S^{(c)} = \frac{1}{k} x_t^\top \sum_{\tau < t} (x_\tau \otimes v_\tau^*), \quad (12)$$

which contains no dependence on g^w whatsoever. The output equals the ungated single-channel form scaled by $1/k$. Consequently

$$\frac{\partial \mathcal{L}}{\partial W_g^w} = 0 \quad \text{identically, whenever the read gate is uniform.} \quad (13)$$

We verified this numerically as CHECK 5: four wildly different near-one-hot write routings (mean commitment 1.0000) evaluated under a uniform read produce outputs identical to 9.4×10^{-16} , all equal to the ungated single-channel form with scores scaled by $1/k$ to 0.0×10^0 .

4.3 The deadlock

A committed write is *invisible* to a uniform read. Training begins with a near-uniform read gate, so the write gate starts with no gradient signal about how to route.

The converse arm is equally exact. When write routing is uniform, every channel accumulates the same $1/k$ share of every write, so all channels hold identical content $S^{(c)} = \bar{S}$, and then

$$a_t^* = \sum_{c=1}^k g_t^r[c] (x_t^\top \bar{S}) = (x_t^\top \bar{S}) \underbrace{\sum_{c=1}^k g_t^r[c]}_{=1} = x_t^\top \bar{S}, \quad (14)$$

independent of g^r . The read gate is exactly as invisible to undifferentiated channels as a committed write is to a uniform read. Both gradients vanish at the symmetric point, and neither side can move first.

This explains why hardening the write leaves the collapse untouched: it changes *how* the write commits, not *whether the read can see it*. It also explains the diffuse, non-degenerate character of the failure in Experiment 5—with no gradient pressure in either direction, the write routing simply drifts, which is exactly what the $[0.335, 0.665] / [0.448, 0.552]$ endpoint looks like.

4.4 Correction to Revision 1

Revision 1 of this report attributed the collapse to the Hebbian write making permissive routing *locally optimal*: spreading writes populates more total usable state, so mixing reads better, and “the router would need to withhold writes from all-but-one channel to create separation, but withholding writes is precisely what the plasticity-driven loss penalizes.”

The second clause is wrong, and equation (11) shows why. With a normalized gate, the router *cannot* withhold writes in any sense the loss can detect: withholding from one channel necessarily gives that mass to another, and the channel sum is conserved. The problem is not that withholding is penalized. The problem is that routing is *unobservable* through an unseparated read, so it is neither penalized nor rewarded—it is simply invisible.

Revision 1 was right that a deadlock exists, but wrong about its location. It follows that Revision 1’s own constructive proposal—decouple the write from the routing—does not work, and Experiment 5 measures it at zero lift. The correct prescription is the opposite one: *any viable intervention must move the read gate*, not the write gate alone. We state this as a testable prediction rather than a result, since we have not yet run it.

4.5 What the cancellation does not explain

The cancellation is exact at the *symmetric point*—a uniform read gate, or undifferentiated channels—which is the regime in which training begins. It therefore accounts for why separation never emerges from a standard initialization, the observed failure in Experiments 1–3 and 5.

It does not, by itself, account for Experiment 4’s erosion result. There the gate was initialized to the perfect separated solution, so the read gate was *not* uniform and the channels were *not* undifferentiated: equations (12) and (14) do not apply, a non-zero gradient was available, and something nonetheless drove the configuration back toward mixing. The accumulation argument of Revision 1—that spreading writes populates more total usable state, so mixing reads better—may well remain the operative force away from the symmetric point, even though it is exactly wrong *at* that point. We have not separated these possibilities. The stability of the separated solution under a non-uniform read is the clearest open question left by this work.

5 Verification for the asymmetric variant

Hard write routing has no unconditional parallel form, so the machine-precision parallel/recurrent identity used for Experiments 1–4 does not apply to Experiment 5. We replaced it with five numeric checks, all passing.

Check	Result
1. $k=1$ reduces exactly to single-channel BDH, across $\tau \in [0.1, 50]$	0.00×10^0
2. High temperature converges to the uniform-gate baseline as $O(1/\tau)$	4.4×10^{-4} at $\tau=10^4$; 6.2×10^{-8} at $\tau=10^8$
3. Gumbel-max property: committed channel distributed as the soft gate (40k draws vs. true $[0.4700, 0.5300]$)	max dev. 0.0047 (tol. 0.0200)
4. With sampled noise fixed, the parallel/recurrent identity is restored	1.55×10^{-15}
5. Write routing cancels under a uniform read (four near-one-hot routings)	9.4×10^{-16}

Table 5: Verification checks for the asymmetric-routing implementation. Check 1 holds because a softmax over a single logit is identically 1. Check 4 pins the gate orientation—read on the query row, write on the source column—which a purely stochastic implementation would otherwise leave ambiguous. Check 5 is the cancellation identity of Section 4.

One correction to the original experimental specification is worth recording. We had expected the high-temperature limit of the Concrete relaxation to approach $\text{softmax}(\text{logits})$; it does not. The relaxation flattens toward the uniform distribution $[1/k, \dots, 1/k]$ as $\tau \rightarrow \infty$. This turned out to be the more useful limit: since $g^r \cdot [1/k, \dots, 1/k] = 1/k$ for *any* read gate, and the uniform-gate floor’s $g \cdot g$ is also $1/k$, the high- τ asymmetric model converges to the exact floor condition of Experiment 2—checkable at machine precision, as Check 2 reports. The intended claim, that the hard write is faithful to the soft gate, is true at the level of *which channel is chosen*, and that is Check 3.

6 Conclusion

A context-gated multi-channel Hebbian architecture is expressible and provably capable: a hand-set gate solves a memory-interference task (100%) that a single channel caps at 50%. But end-to-end training never uses that capacity. The gate collapses to permissive routing across soft, auxiliary-loss-augmented, hard-discrete, and write/read-decoupled variants; the separated solution is not a stable fixed point; and no variant captured any part of the available 50-accuracy-point margin.

The cause of that failure to bootstrap is specific, verified, and—to our knowledge—previously undocumented in this form. Because the write gate is a normalized distribution, the channel-summed

state is invariant to write routing, so under a uniform read gate the routing cancels exactly and $\partial\mathcal{L}/\partial W_g^w$ is identically zero. Training begins in a bootstrapping deadlock: the write gate cannot learn to route because the read gate cannot see routing, and the read gate has nothing separated to attend to because the write gate has not routed.

We report this as a negative result in the spirit of saving subsequent effort. The idea of multi-neurotransmitter-like synaptic channels is intuitive and recurs naturally; this report documents, with a ladder of five experiments and an exact mechanistic account, why the straightforward end-to-end realization of it does not work on the tasks studied. We do not claim the result generalizes beyond the small scale, the single architecture family, and the task designs examined here; establishing its scope would require further work. But within that scope the finding is unambiguous and consistently reproduced.

A constructive reading points to where a future attempt would have to differ. The cancellation of equation (12) depends on two properties: that the write gate is normalized, and that the read gate is uniform at initialization. Breaking either should restore a gradient signal. Candidate interventions include an *unnormalized* or sparsity-penalized write gate, so that total written mass is not conserved and withholding becomes observable; a read gate initialized away from uniform, so the write gate has signal from the first step; bounded per-channel capacity (for example a norm constraint or Oja-style decay), so that writes compete rather than accumulate freely; or an error-driven write rule, so that a channel already holding an association stops writing it. Each of these attacks the cancellation directly rather than, as Revision 1 proposed, the write/read coupling—which we have now measured and found not to be the obstacle.

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